

Lab 2: Scanning and Enumeration (Nmap + Metasploit) (Assignment and In Class exercisess)

Lab Objective

In this lab, you will perform scanning and enumeration against multiple systems using Nmap and Metasploit auxiliary scanners. Each target will be scanned independently, allowing you to work with one target VM at a time.

You will:

- Identify live hosts and open ports
- Enumerate services and versions
- Use Nmap NSE scripts
- Use Metasploit auxiliary scanners
- Analyze results and prepare an attack plan (no exploitation)

Lab Environment

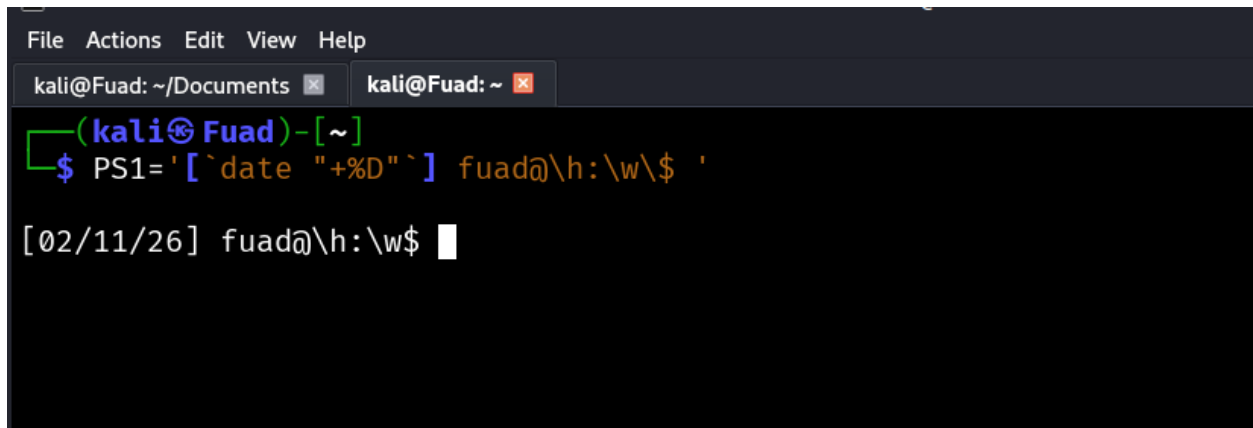
VM	Role
ATTACK01	Kali Linux
MSF01	Metasploitable 2
DC01	Windows Server 2019 (AD)
WS01	Windows 10
Network	VMware Host-Only (VMnet1)

Rules of Engagement

- Do not exploit vulnerabilities
- Follow instructions per VM section
- Capture only the required screenshots

NOTE: For every tabs you open, make sure your name is on the tab using this command. For screenshot tabs you do not add your name, half mark would be deducted

```
PS1='[`date "+%D"`] yourname@\h:\w\$ '
```



The screenshot shows a Kali Linux terminal window with a menu bar (File, Actions, Edit, View, Help) and two tabs: 'kali@Fuad: ~/Documents' and 'kali@Fuad: ~'. The active tab is 'kali@Fuad: ~'. The terminal prompt is '(kali@Fuad)-[~]'. The user has entered the command 'PS1='[`date "+%D"`] fuad@\h:\w\\$ ' and the prompt has changed to '[02/11/26] fuad@\h:\w\$'.

```
(kali@Fuad)-[~]  
$ PS1='[`date "+%D"`] fuad@\h:\w\$ '  
[02/11/26] fuad@\h:\w$
```

Part 1 - Kali Preparation (ATTACK01)

Step 1.1 - Confirm Network Settings

Confirm IP Address on Kali

```
ip a
```

Record:

- Your IP address

This information will be reused for all scans.

Part 2 — Scanning Metasploitable 2 (MSF01)

VM State

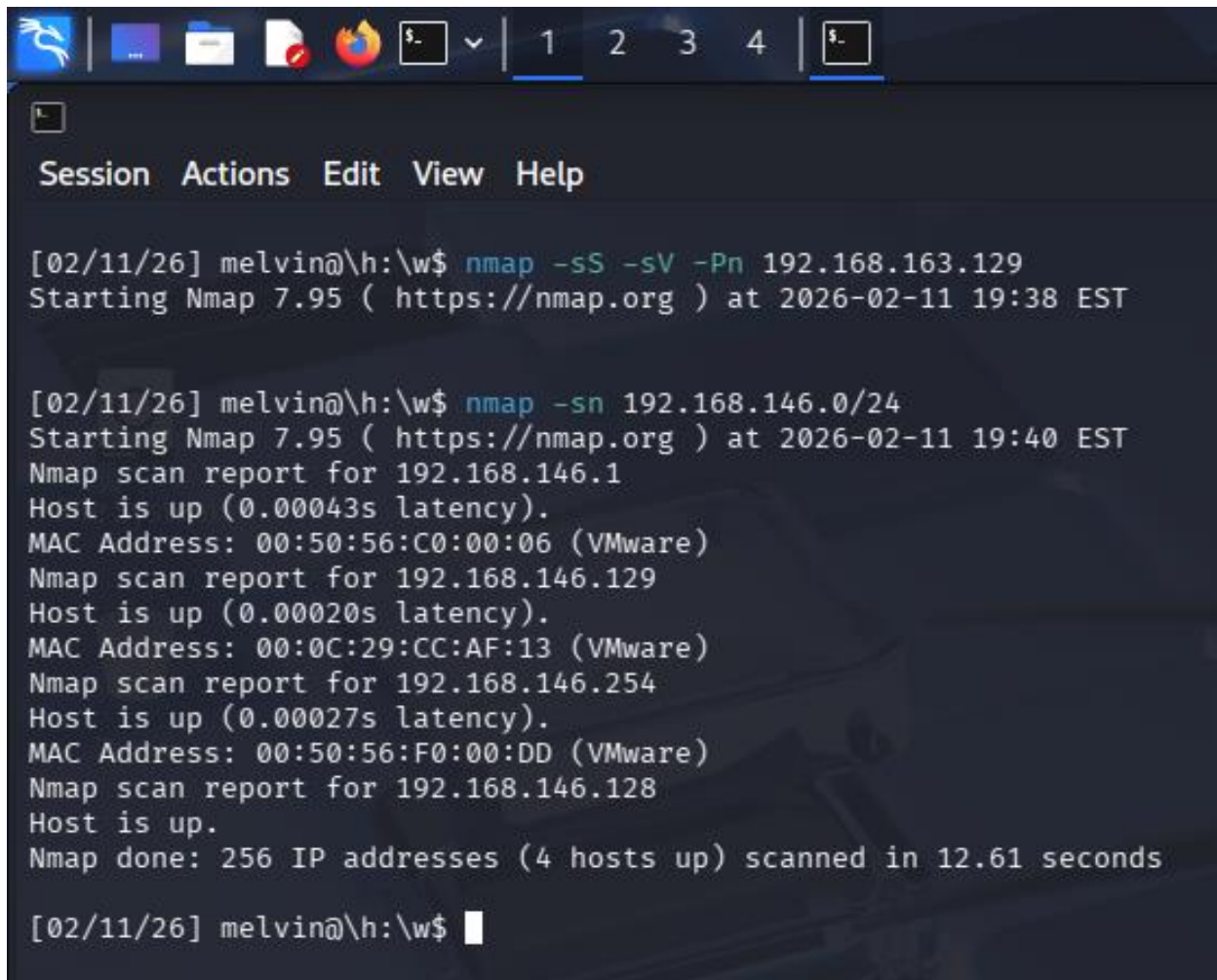
- Power ON: **ATTACK01 + MSF01**

Step 2.1 — Identify Metasploitable IP

```
nmap -sn <SUBNET>/24
```

Record the IP for MSF01.

[Attach your Screenshot]



The screenshot shows a Windows taskbar at the top with icons for File Explorer, Firefox, and a terminal window. The terminal window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The command prompt shows the following text:

```
[02/11/26] melvin@h:\w$ nmap -sS -sV -Pn 192.168.163.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-11 19:38 EST

[02/11/26] melvin@h:\w$ nmap -sn 192.168.146.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-11 19:40 EST
Nmap scan report for 192.168.146.1
Host is up (0.00043s latency).
MAC Address: 00:50:56:C0:00:06 (VMware)
Nmap scan report for 192.168.146.129
Host is up (0.00020s latency).
MAC Address: 00:0C:29:CC:AF:13 (VMware)
Nmap scan report for 192.168.146.254
Host is up (0.00027s latency).
MAC Address: 00:50:56:F0:00:DD (VMware)
Nmap scan report for 192.168.146.128
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 12.61 seconds

[02/11/26] melvin@h:\w$
```

MSF01 IP – 192.168.146.129

Step 2.2 — Full Service Scan (Nmap)

```
nmap -sS -sV -Pn <MSF01_IP>
```

Observe:

- Number of open ports
- Outdated services

Step 2.3 — NSE Vulnerability Scan

```
sudo nmap -sV --script vuln <MSF01_IP>
```

Observe:

- CVEs
- CVSS scores
- Exploit references

[Attach your Screenshot]

```
[02/11/26] melvin@h:\w$ sudo nmap -sV --script "vuln" 192.168.146.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-11 19:48 EST
Stats: 0:00:56 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 95.65% done; ETC: 19:49 (0:00:02 remaining)
Stats: 0:03:26 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.00% done; ETC: 19:51 (0:00:00 remaining)
Stats: 0:04:58 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.66% done; ETC: 19:53 (0:00:00 remaining)
Nmap scan report for 192.168.146.129
Host is up (0.0020s latency).
Not shown: 777 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp           vsftpd 2.3.4
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|     vsFTPD version 2.3.4 backdoor
|       State: VULNERABLE (Exploitable)
|       IDs:   BID:48539   CVE:CVE-2011-2523
|       vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|       Disclosure date: 2011-07-03
|       Exploit results:
|         Shell command: id
|         Results: uid=0(root) gid=0(root)
|       References:
|         https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|         https://www.securityfocus.com/bid/48539
|         http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|         https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|   vulners:
|     vsftpd 2.3.4:
|       PACKETSTORM:162145      10.0      https://vulners.com/packetstorm/PACKETSTORM:162145 *EXPLOIT*
|       EDB-ID:49757      10.0      https://vulners.com/exploitdb/EDB-ID:49757 *EXPLOIT*
|       E9B0AEBB-5138-50BF-8922-2D87E3C046DD      10.0      https://vulners.com/githubexploit/E9B0AEBB-5138-50BF-8922-2D87E3C046DD *EXPLOIT*
|       CVE-2011-2523      10.0      https://vulners.com/cve/CVE-2011-2523
|       CNVD-2020-46837      10.0      https://vulners.com/cnvd/CNVD-2020-46837
|       CC3F6C15-182F-53F6-A5CC-812D37F1F047      10.0      https://vulners.com/githubexploit/CC3F6C15-182F-53F6-A5CC-812D37F1F047 *EXPLOIT*
|       A4185EAD-1A4C-56A6-97C6-1C58A1CF1E3B      10.0      https://vulners.com/githubexploit/A4185EAD-1A4C-56A6-97C6-1C58A1CF1E3B *EXPLOIT*
|       817CD8FE-87C4-5FE8-B9B2-8CC64333A3F3      10.0      https://vulners.com/githubexploit/817CD8FE-87C4-5FE8-B9B2-8CC64333A3F3 *EXPLOIT*
|       63A5C9A7-C241-5E83-9EE6-ABAB44BDD270      10.0      https://vulners.com/githubexploit/63A5C9A7-C241-5E83-9EE6-ABAB44BDD270 *EXPLOIT*
|       5F4BCEDE-77DF-5D54-851A-0AE8B76458D9      10.0      https://vulners.com/githubexploit/5F4BCEDE-77DF-5D54-851A-0AE8B76458D9 *EXPLOIT*
|       50580586-73C4-5097-81CA-546D6591DF44      10.0      https://vulners.com/githubexploit/50580586-73C4-5097-81CA-546D6591DF44 *EXPLOIT*
|       1337DAY-ID-36095      9.8      https://vulners.com/zdt/1337DAY-ID-36095 *EXPLOIT*
22/tcp    open  ssh           OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| vulners:
|   cpe:/a:openbsd:openssh:4.7p1:
|     DF059135-2CF5-5441-8F22-E6EF1DEE5F6E      10.0      https://vulners.com/itee/DF059135-2CF5-5441-8F22-E6EF1DEE5F6E *EXPLOIT*
|     PACKETSTORM:173661      9.8      https://vulners.com/packetstorm/PACKETSTORM:173661 *EXPLOIT*
|     F0979183-AE88-53B4-86CF-3AF0523F3807      9.8      https://vulners.com/githubexploit/F0979183-AE88-53B4-86CF-3AF0523F3807 *EXPLOIT*
|     CVE-2023-38408      9.8      https://vulners.com/cve/CVE-2023-38408
|     CVE-2016-1908      9.8      https://vulners.com/cve/CVE-2016-1908
|     B8190CDB-3EB9-5631-9828-8064A1575B23      9.8      https://vulners.com/githubexploit/B8190CDB-3EB9-5631-9828-8064A1575B23 *EXPLOIT*
|     8FC9C5AB-3968-5F3C-825E-E8DB5379A623      9.8      https://vulners.com/githubexploit/8FC9C5AB-3968-5F3C-825E-E8DB5379A623 *EXPLOIT*
|     8AD01159-548E-546E-AA87-2DE89F3927EC      9.8      https://vulners.com/githubexploit/8AD01159-548E-546E-AA87-2DE89F3927EC *EXPLOIT*
|     6192C35D-F78B-5C0A-AB8D-9826A79A5320      9.8      https://vulners.com/githubexploit/6192C35D-F78B-5C0A-AB8D-9826A79A5320 *EXPLOIT*
|     2227729D-6700-5C8F-8930-1EEAFD4B9FF0      9.8      https://vulners.com/githubexploit/2227729D-6700-5C8F-8930-1EEAFD4B9FF0 *EXPLOIT*
|     0221525F-07F5-5790-912D-F4B9E2D1B587      9.8      https://vulners.com/githubexploit/0221525F-07F5-5790-912D-F4B9E2D1B587 *EXPLOIT*
|     CVE-2015-5600      8.5      https://vulners.com/cve/CVE-2015-5600
```

Step 2.4 — Metasploit Auxiliary Scanning

Launch Metasploit: `msfconsole` and change the console prompt:

```
(kali@Fuad)-[~/Documents]  
$ msfconsole  
Metasploit tip: Open an interactive Ruby terminal with irb  
[*] Starting the Metasploit Framework console... \
```

Msf6> set PROMPT %yel%L %grn%T %grnyourfirstname

```
msf6 > set PROMPT %yel%L %grn%T %grnfuad  
PROMPT => %L %T fuad  
192.168.227.188 2026-02-11 00:11:33 fuad > █
```

Run the following:

```
use auxiliary/scanner/portscan/tcp
```

```
set RHOSTS <MSF01_IP>
```

```
run
```

```
msf6 > set PROMPT %yel%L %grn%T %grnfuad  
PROMPT => %L %T fuad  
192.168.227.188 2026-02-11 00:11:33 fuad > use auxiliary/scanner/portscan/tcp  
192.168.227.188 2026-02-11 00:29:13 fuad auxiliary(scanner/portscan/tcp) > set RHOSTS 192.168.227.203  
RHOSTS => 192.168.227.203  
192.168.227.188 2026-02-11 00:29:18 fuad auxiliary(scanner/portscan/tcp) > run  
[+] 192.168.227.203: - 192.168.227.203:21 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:22 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:23 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:25 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:53 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:80 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:111 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:139 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:445 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:512 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:513 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:514 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:599 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:1524 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:2049 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:2121 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:3306 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:3632 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:5432 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:5900 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:6000 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:6667 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:6697 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:8009 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:8180 - TCP OPEN  
[+] 192.168.227.203: - 192.168.227.203:8787 - TCP OPEN  
[*] 192.168.227.203: - Scanned 1 of 1 hosts (100% complete)  
[*] Auxiliary module execution completed  
192.168.227.188 2026-02-11 00:29:43 fuad auxiliary(scanner/portscan/tcp) > █
```

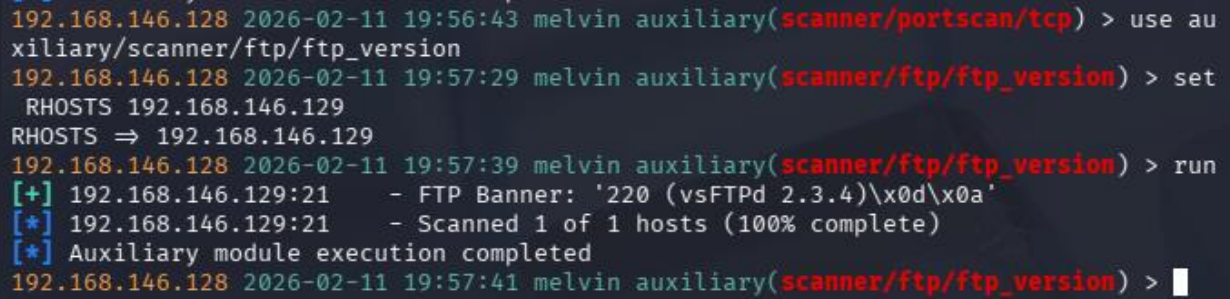

[Attach your Screenshot]

```
kali@kali: ~  
Session Actions Edit View Help  
%%  
%%  
%%  
Press SPACE BAR to continue  
  
=[ metasploit v6.4.99-dev ]  
+ -- --[ 2,572 exploits - 1,317 auxiliary - 1,680 payloads ]  
+ -- --[ 432 post - 49 encoders - 13 nops - 9 evasion ]  
  
Metasploit Documentation: https://docs.metasploit.com/  
The Metasploit Framework is a Rapid7 Open Source Project  
  
msf > set PROMPT %yel%L %grn%T %grnmelvin  
PROMPT => %L %T melvin  
192.168.146.128 2026-02-11 19:55:21 melvin > use auxiliary/scanner/portscan/t  
cp  
192.168.146.128 2026-02-11 19:55:37 melvin auxiliary( - this by opening connec  
scanner/portscan/tcp) > set RHOSTS 192.168.146.128  
RHOSTS => 192.168.146.128  
192.168.146.128 2026-02-11 19:55:48 melvin auxiliary(  
scanner/portscan/tcp) > run  
[*] 192.168.146.128 - Scanned 1 of 1 hosts (100% complete)  
[*] Auxiliary module execution completed  
192.168.146.128 2026-02-11 19:56:04 melvin auxiliary(scanner/portscan/tcp) > set RH  
OSTS 192.168.146.129  
RHOSTS => 192.168.146.129  
192.168.146.128 2026-02-11 19:56:30 melvin auxiliary(scanner/portscan/tcp) > run  
[+] 192.168.146.129 - 192.168.146.129:21 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:22 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:23 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:25 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:53 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:80 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:111 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:139 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:445 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:512 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:513 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:514 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:1099 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:1524 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:2049 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:2121 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:3306 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:3632 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:5432 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:5900 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:6000 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:6667 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:6697 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:8009 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:8180 - TCP OPEN  
[+] 192.168.146.129 - 192.168.146.129:8787 - TCP OPEN  
[*] 192.168.146.129 - Scanned 1 of 1 hosts (100% complete)  
[*] Auxiliary module execution completed  
192.168.146.128 2026-02-11 19:56:43 melvin auxiliary(scanner/portscan/tcp) > 
```


Then enumerate services:

```
use auxiliary/scanner/ftp/ftp_version  
set RHOSTS <MSF01_IP>  
run
```

[Attach your Screenshot]



```
192.168.146.128 2026-02-11 19:56:43 melvin auxiliary(scanner/portscan/tcp) > use au  
xiliary/scanner/ftp/ftp_version  
192.168.146.128 2026-02-11 19:57:29 melvin auxiliary(scanner/ftp/ftp_version) > set  
RHOSTS 192.168.146.129  
RHOSTS => 192.168.146.129  
192.168.146.128 2026-02-11 19:57:39 melvin auxiliary(scanner/ftp/ftp_version) > run  
[+] 192.168.146.129:21 - FTP Banner: '220 (vsFTPd 2.3.4)\x0d\x0a'  
[*] 192.168.146.129:21 - Scanned 1 of 1 hosts (100% complete)  
[*] Auxiliary module execution completed  
192.168.146.128 2026-02-11 19:57:41 melvin auxiliary(scanner/ftp/ftp_version) > 
```

Screenshot Requirements (Metasploitable)

- Nmap service scan
- NSE vuln output
- Metasploit auxiliary scan results

Part 3 — Scanning Windows Server (DC01 – AD)

VM State

- Power ON: **ATTACK01 + DC01**

Step 3.1 — Identify DC01 IP

```
nmap -sn <SUBNET>/24
```

Step 3.2 — Targeted AD Port Scan

```
nmap -sS -sV -Pn -p 53,88,135,139,389,445,464,636,3268,3269 <DC01_IP>
```

Observe:

- DNS
- Kerberos
- LDAP
- SMB

```
[02/11/26] fuad@h:\w$ nmap -sS -sV -Pn -p 53,88,135,139,389,445,464,636,3268,3269 192.168.227.216
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-11 00:33 EST
Nmap scan report for 192.168.227.216
Host is up (0.00091s latency).

PORT      STATE SERVICE          VERSION
53/tcp    open  domain           Simple DNS Plus
88/tcp    open  kerberos-sec     Microsoft Windows Kerberos (server time: 2026-02-11 05:33:49Z)
135/tcp    open  msrpc            Microsoft Windows RPC
139/tcp    open  netbios-ssn     Microsoft Windows netbios-ssn
389/tcp    open  ldap             Microsoft Windows Active Directory LDAP (Domain: DCFuad.org0., Site: Default-First-Site-Name)
445/tcp    open  microsoft-ds?
464/tcp    open  kpasswd5?
636/tcp    open  tcpwrapped
3268/tcp   open  ldap             Microsoft Windows Active Directory LDAP (Domain: DCFuad.org0., Site: Default-First-Site-Name)
3269/tcp   open  tcpwrapped
MAC Address: 00:0C:29:B9:0F:C2 (VMware)
Service Info: Host: FUAD-DC01; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.49 seconds

[02/11/26] fuad@h:\w$
```



[Attach your Screenshot]

```
[02/11/26] melvin@h:\w$ nmap -sS -sV -Pn -p 53,88,135,139,389,445,464,636,3268,3269 192.168.146.130
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-11 21:18 EST
Stats: 0:00:00 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 10.00% done; ETC: 21:18 (0:00:00 remaining)
Stats: 0:23:11 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 20.00% done; ETC: 23:14 (1:32:44 remaining)
Nmap scan report for 192.168.146.130
Host is up (0.0013s latency).

PORT      STATE SERVICE      VERSION
53/tcp    open  domain       Simple DNS Plus
88/tcp    open  kerberos-sec Microsoft Windows Kerberos (server time: 2026-02-12 02:18:41Z)
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
389/tcp   open  ldap         Microsoft Windows Active Directory LDAP (Domain: DCMelvin.org0., Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds Microsoft Windows Server 2008 R2 - 2012 microsoft-ds (workgroup: DCMelvin)
464/tcp   open  kpasswd5?
636/tcp   open  tcpwrapped
3268/tcp  open  ldap         Microsoft Windows Active Directory LDAP (Domain: DCMelvin.org0., Site: Default-First-Site-Name)
3269/tcp  open  tcpwrapped
MAC Address: 00:0C:29:25:CB:8F (VMware)
Service Info: Host: MELVIN-DC01; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1391.59 seconds
```

Step 3.3 — SMB Enumeration

```
smbclient -L //<DC01_IP> -N
```

```
[02/11/26] fuad@h:\w$ smbclient -L //192.168.227.216 -N
```

Sharename	Type	Comment
ADMIN\$	Disk	Remote Admin
C\$	Disk	Default share
Common	Disk	
IPC\$	IPC	Remote IPC
NETLOGON	Disk	Logon server share
SYSVOL	Disk	Logon server share

```
Reconnecting with SMB1 for workgroup listing.  
do_connect: Connection to 192.168.227.216 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)  
Unable to connect with SMB1 -- no workgroup available
```

Sample Screenshot

```
[02/11/26] fuad@h:\w$
```

[Attach your Screenshot]

```
[02/11/26] melvin@h:\w$ smbclient -L //192.168.163.130 -N  
do_connect: Connection to 192.168.163.130 failed (Error NT_STATUS_IO_TIMEOUT)
```

Step 3.4 — Metasploit Auxiliary AD Scanning

```
use auxiliary/scanner/smb/smb_version

set RHOSTS <DC01_IP>

run
```

```
192.168.227.188 2026-02-11 00:31:12 fuad auxiliary(scanner/ftp/ftp_version) > use auxiliary/scanner/smb/smb_version
192.168.227.188 2026-02-11 00:36:10 fuad auxiliary(scanner/smb/smb_version) > set RHOSTS 192.168.227.216
RHOSTS => 192.168.227.216
192.168.227.188 2026-02-11 00:36:21 fuad auxiliary(scanner/smb/smb_version) > run
[*] 192.168.227.216:445 - SMB Detected (versions:2, 3) (preferred dialect:SMB 3.1.1) (compression capabilities:) (encryption capabilities:AES-128-GCM) (signatures:required) (guid:{295d9936-26b6-4057-bf64-5ca178b848cc}) (authentication domain:DCFuad)
[+] 192.168.227.216:445 - Host is running Version 10.0.17763 (likely Windows 10 version 1809/Windows Server 2019)
[*] 192.168.227.216: - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
192.168.227.188 2026-02-11 00:36:25 fuad auxiliary(scanner/smb/smb_version) > 
```

Sample Screenshot

Run this other command and add yu

```
use auxiliary/scanner/netbios/nbname

set RHOSTS <DC01_IP>

run
```

[Attach your Screenshot]

```
192.168.146.128 2026-02-11 20:45:09 melvin auxiliary(scanner/netbios/nbname) > set RHOSTS 192.168.146.130
RHOSTS => 192.168.146.130
192.168.146.128 2026-02-11 20:45:27 melvin auxiliary(scanner/netbios/nbname) > run
[*] Sending NetBIOS requests to 192.168.146.130→192.168.146.130 (1 hosts)
[+] 192.168.146.130 [MELVIN-DC01] OS:Windows Names:(MELVIN-DC01, DCMELVIN, __MSBROWSE__) Addresses:(192.168.146.130) Mac:00:0c:29:25:cb:8f Virtual Machine
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
192.168.146.128 2026-02-11 20:46:36 melvin auxiliary(scanner/netbios/nbname) > ss
```

Part 4 — Scanning Windows 10 (WS01)

VM State

- Power ON: **ATTACK01 + WS01**

Step 4.1 — Identify WS01 IP

```
nmap -sn <SUBNET>/24
```

[Attach your Screenshot]

```
[02/11/26] melvin@h:\w$ nmap -sn 192.168.146.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-11 20:10 EST
Nmap scan report for 192.168.146.1
Host is up (0.00016s latency).
MAC Address: 00:50:56:C0:00:06 (VMware)
Nmap scan report for 192.168.146.2
Host is up (0.000092s latency).
MAC Address: 00:50:56:F4:6E:84 (VMware)
Nmap scan report for 192.168.146.131
Host is up (0.00021s latency).
MAC Address: 00:0C:29:EE:4C:90 (VMware)
Nmap scan report for 192.168.146.254
Host is up (0.00014s latency).
MAC Address: 00:50:56:F0:00:DD (VMware)
Nmap scan report for 192.168.146.128
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 1.92 seconds
[02/11/26] melvin@h:\w$
```

Step 4.2 — Service Scan

```
nmap -sS -sV -Pn -p 135,139,445,3389,5985,5986 <WS01_IP>
```

[Attach your Screenshot]

```
[02/11/26] melvin@h:\w$ nmap -sS -sV -Pn -p 53,88,135,139,389,445,464,636,3268,3269 192.168.146.130
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-11 20:47 EST
Nmap scan report for 192.168.146.130
Host is up (0.00065s latency).

PORT      STATE SERVICE      VERSION
53/tcp    open  domain       Simple DNS Plus
88/tcp    open  kerberos-sec Microsoft Windows Kerberos (server time: 2026-02-12 01:47:31Z)
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
389/tcp   open  ldap         Microsoft Windows Active Directory LDAP (Domain: DCMelvin.org0., Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds Microsoft Windows Server 2008 R2 - 2012 microsoft-ds (workgroup: DCMelvin)
464/tcp   open  kpasswd5?
636/tcp   open  tcpwrapped
3268/tcp  open  ldap         Microsoft Windows Active Directory LDAP (Domain: DCMelvin.org0., Site: Default-First-Site-Name)
3269/tcp  open  tcpwrapped
MAC Address: 00:0C:29:25:CB:8F (VMware)
Service Info: Host: MELVIN-DC01; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.50 seconds

[02/11/26] melvin@h:\w$
```

Step 4.3 — Metasploit Auxiliary Scan

```
use auxiliary/scanner/smb/smb_version
```

```
set RHOSTS <WS01_IP>
```

```
run
```

[Attach your Screenshot]

```
192.168.146.128 2026-02-11 20:48:34 melvin auxiliary(scanner/smb/smb_version) > run
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.23/lib/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator '+'
[*] 192.168.146.131:445 - SMB Detected (versions:2, 3) (preferred dialect:SMB 3.1.1) (compression capabilities:LZNT1) (encryption capabilities:AES-128-GCM)
[*] 192.168.146.131:445 - Host is running Version 10.0.19041 (likely Windows 10 version 2004)
[*] 192.168.146.131 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
```

```
use auxiliary/scanner/rdp/rdp_scanner
```

```
set RHOSTS <WS01_IP>
```

```
run
```


Part 5 — Analysis & Enumeration Summary

Answer the following (½–1 page):

1. Which system exposed the largest attack surface?

Metasploitable 2 (MSF01) exposed the largest attack surface. After running the NSE vulnerability scan, it produced the most results showing many common vulnerabilities and exploits.

2. Which services appear most often in Windows/AD environments?

Services that appear most frequently in Windows or Active Directory environments that are attractive to attackers are Server Message Block (SMB), Remote Desktop (RDP), Windows Remote Management. Other services are Active Directory Domain Services (ADDS), DNS Server, LDAP, and Kerberos.

3. How do Metasploit auxiliary scans complement Nmap?

You can use Nmap to quickly scan ports or services, then use the auxiliary scanners of Metasploit to scan for specific services. Nmap is effective for scanning a large network while Metasploit is better for targeting for specific ports or services. Metasploit is also effective for detecting vulnerabilities and exploits on devices to achieve greater depth in penetration testing.

4. Which target would you attack first and why? (No exploitation)

I would attack Metasploitable (MSF01) because it produced many vulnerabilities from the Nmap NSE vulnerability scan. The Domain Controller (DC01) and the Windows Server (WS01) compared to the Metasploitable device may have more secure environments with less vulnerabilities that would be more difficult to attack as a malicious actor or penetration tester. The Metasploitable image is setup with vulnerabilities to practice attack the system anyways.

Lab Completion Checklist

You are finished when:

- Each VM was scanned independently
- Both Nmap and Metasploit scanners were used
- Enumeration results were analyzed
- Exactly **10 screenshots** were submitted

Notes for Students

- **Screenshots must be clear and readable**
- **Screenshots must show your own work**
- **Do not crop out URLs, commands, or results**
- **Missing screenshots will result in an incomplete grades**